



Improving I/O Performances of HPC Application Gysela

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**TECHNICAL
REPORT**

N° XXXX

MONTH YEAR

Project-Teams TADaaM



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Technical Report n° XXXX — MONTH YEAR — 4 pages

Abstract: In modern High Performances Computing's (HPC) systems, Input/Output (I/O) performances are often a limiting factor for large scale scientific simulations. Since supercomputer power and data volume keep increasing, I/O performance optimization becomes essential in order to efficiently exploit available resources. In this context, scientific applications are largely based on Parallel File System (PFS) enabling the access at the same time to very large amount of data. However, the results achieved highly depend of many configuration's parameters as well as grounds for accessing the data used by applications. Understanding and Optimizing those interactions constitutes a crucial issue for improving HPC simulations global efficiency.

This work is part of the optimization of I/O performances of the application called Gysela, a simulation code developed by CEA for the study of plasma's turbulences in Tokamak, for the study of nuclear fusion. Those simulations produce large amount of data and needed regular checkpoints writing, that allow to save the state of the simulation it can resume it in case of an interruption, whether it is due to a hardware failure or a limitation on the allotted processing time on supercomputers.

Key-words: HPC, Parallel I/O, Performances Tuning, Parallel File Systems

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1 Introduction

The growing gap between computational performance and I/O capabilities in modern HPC systems has become a critical concern for large-scale scientific simulations. As supercomputers approach exascale, applications generate ever-increasing volumes of data, making efficient data management essential for both performance and fault tolerance. Among the various I/O patterns found in scientific codes, checkpointing—the periodic saving of simulation state to enable job restart after failures or time limits—is particularly demanding, as it involves writing large amounts of data in a short time window.

2 Second section

Text for the second section. This is closely related to the text in section 1 on page 4.

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Publisher
Inria
Domaine de Voluceau - Rocquencourt
BP 105 - 78153 Le Chesnay Cedex
inria.fr

ISSN 0249-0803